

Article

Blackcurrants - a Kiwi example of a positive food

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If an illustration is needed of what Plant & Food Research call “Positive Foods”, look no further than the humble Kiwi blackcurrant – a berry that has health, nutrition and sustainability implications, not to mention Olympic significance when it comes to exercise and recovery.





Plant & Food Research is the Crown Research Institute which uses science to boost New Zealand's food sectors and General Manager Science, Dr Jocelyn Eason, says positive foods are a key focus of their work: "When we talk about positive foods, we mean good for people and good for planet so they're nutritious and they have a positive environmental footprint.

"Sustainability is a holistic issue for us. It's about the environment, it's about people's welfare, it's about people's health, it's about job creation. Sustainability has social impacts as well as environmental impacts so when we talk about positive foods, we're talking about sustainability across all of those different aspects."



It's also about designing and directing different foods at different people for different reasons – like the New Zealand blackcurrant. It's a remarkable fruit which has, up to now, escaped the attention of many New Zealanders – but is building an international reputation as a food which supports exercise performance and recovery, immunity and even cognitive performance.

That's a lot of claims for a modest little berry. But Dr Dominic Lomiwes, Science Team Leader says clinical trials have established there is ample scientific evidence and that blackcurrants demonstrate the application of positive foods at work.

Perhaps the most interesting findings stem from a meta-analysis conducted by scientists from the University of Auckland and Plant & Food Research, published in the Journal of the International Society of Sports Nutrition in 2020. It concluded that the unique



According to this meta-analysis, which included nine clinical studies, consuming New Zealand blackcurrants could improve sports performance by 0.45 per cent compared to a placebo.

That may not seem like much on the surface until you realise the average difference between gold and silver at the 2016 Rio Olympics was 0.54 per cent according to research by Swiss-based open science platform Frontiers (which looked at finishing times across a range of different sports).

So the boost provided by a natural, plant-based performance enhancer is significant. So, what other exercise and health benefits may there be?

“When we exercise, we produce a lot of physiological stress on our bodies,” says Lomiwes. “Blackcurrants promote our body's natural antioxidant systems - to help recovery - and that has implications in terms of health. Post-exercise, the hypothesis is that we are left more susceptible to colds and flu – but if you consume blackcurrants in a timely manner prior to exercise, it aids recovery and helps to close that window of susceptibility.

“Blackcurrants are also often regarded as good for cognitive performance,” says Lomiwes. “They can play a role in inhibiting the enzyme that breaks down dopamine, one of the neurotransmitters in the brain that makes you feel good and can aid performance.”

Eason says Plant & Food Research is doing a whole range of different things in the positive food space: “We’re understanding the environmental footprints (including carbon, nitrogen, phosphorous and water) of our foods. We're understanding the nutrition of the

consumer and we're designing our production systems to use all of the bio-resource and limit waste."

Seafood is an area where Plant & Food Research is helping the sector to become more sustainable.

"Seafood is a great part of a balanced diet," says Helen Mussely, General Manager Science Seafood Technologies. "If we're thinking about positive foods, it's healthy and nutritious for us. There's evidence it's great for brain health and for heart health.

"It's a pretty key part of the kind of diet we should be aiming for, as long as we can produce that seafood in a responsible and sustainable manner. Global consumers seem to agree with that; it's very sought after, especially seafood from Aotearoa."



Plant & Food Research scientists are looking not only at the health benefits of seafood – Lomiwes, for example, is leading a trial looking at the impact of New Zealand Greenshell mussels on exercise recovery - but also how it's sustainably produced.

landing the catch in a better condition.

Next on the list is looking at how to make the most of the ocean by building mobile fish farms that could expand New Zealand aquaculture out into the sea, helping support the Government's targets of a \$3 billion sector with 6000 regional jobs by 2035.

To complement this, Plant & Food Research scientists are also looking at the potential for cellular aquaculture, growing cultured fish meat in a lab environment. This would allow New Zealand to produce seafood products from a factory as well as the ocean.

It's all about choice, Eason says. And consumers agree, with a third of New Zealanders reportedly eating less meat (Hungry for Plant-based Foods report, Food Frontier, 2019) for both health and environmental reasons.

"We have to produce foods that allow consumers to be flexible in their dietary choices," Eason says. "It's not just about meat, it's about nutrition, so where are you getting that nutrition from? Are you getting it from an animal or are you getting it from a plant? Is it sustainable or not? For our agricultural systems to work really well, it's actually quite good to have animals in them.

"At my age and stage my dietary needs are different from others; protein is perhaps less important for me than it is for young children who are highly energetic and growing rapidly. Dairy is another really important nutritional food for infants – but we don't need as much of it at other times of our life when we can eat a variety of other foods to support our health and wellbeing."

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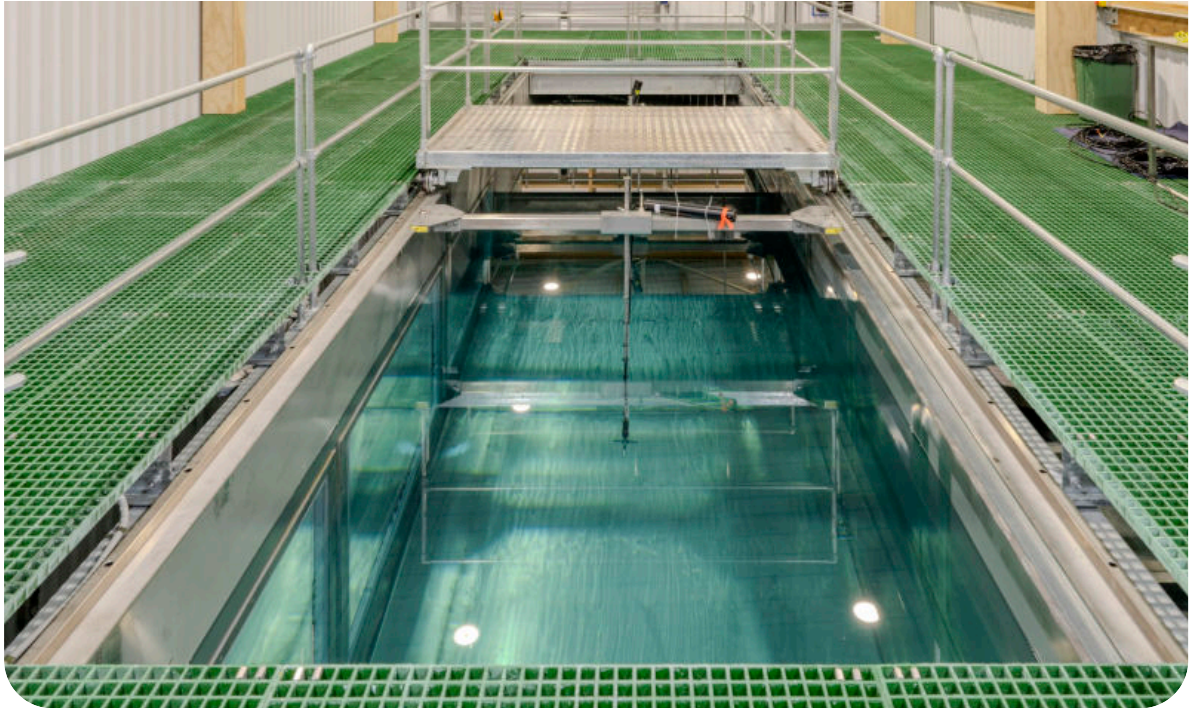
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I am the Science Team Leader for Immune Health and Physical Performance within the Food Innovation Portfolio, based in Palmerston North. Our team conducts in vitro, animal and human clinical...

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